

Intro to Econ 628

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UBC

September 9, 2026

Why is this course important?

Technology and Big Data Are Changing Economics: Mining Text to Track Methods

Janet Currie, Henrik Kleven, and Esmée Zwiers

NBER Working Paper No. 26715

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JEL No. A0,B0,C0,H0,I0,J0,L0

ABSTRACT

The last 40 years have seen huge innovations in computing technology and data availability. Data derived from millions of administrative records or by using (as we do) new methods of data generation such as text mining are now common. New data often requires new methods, which in turn can inspire new data collection. If history is any guide, some methods will stick and others will prove to be a flash in the pan. However, the larger trends towards demanding greater credibility and transparency from researchers in applied economics and a “collage” approach to assembling evidence will likely continue.

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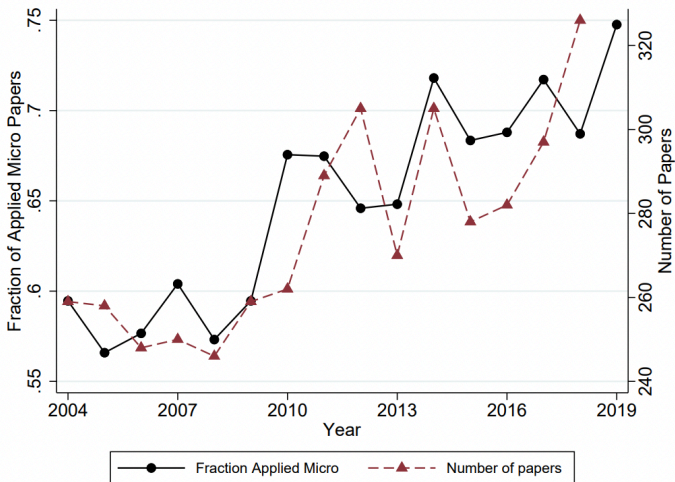
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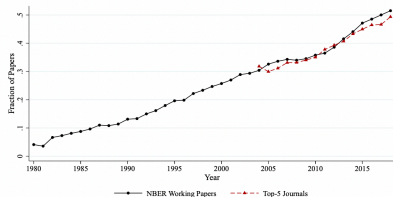
Figure I: Fraction of Applied Microeconomics Articles in Top-5 Journals



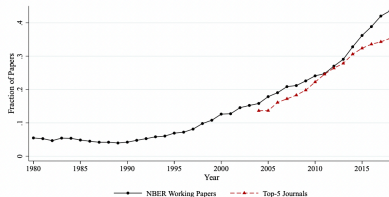
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Figure II: The Credibility Revolution

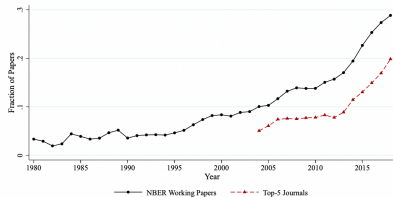
A: Identification



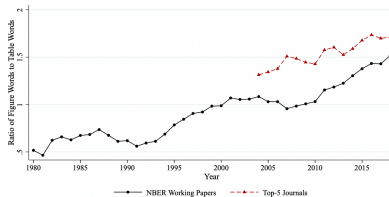
B: All Experimental and Quasi-Experimental Methods



C: Administrative Data



D: Graphical Revolution

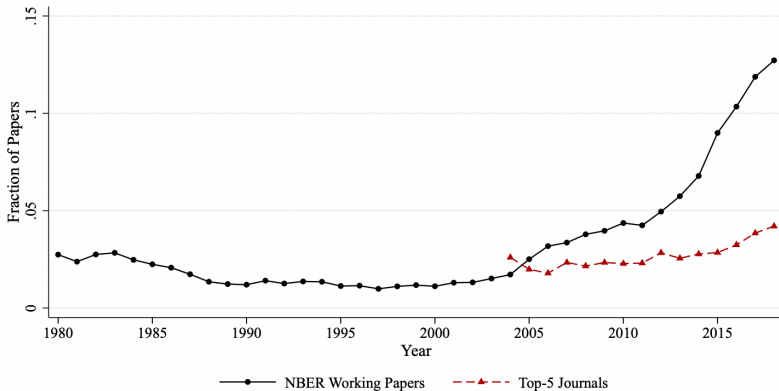


Notes: This figure shows different dimensions of the “credibility revolution” in economics: Identification (Panel A), All Experimental and Quasi-Experimental Methods (Panel B), Administrative Data (Panel C), and the Graphical Revolution (Panel D). Panel D shows the ratio of the number of ‘Figure’ terms mentioned to the number of ‘Table’ terms mentioned. See Table A.1 for a list of terms. The series show 5-year moving averages.

Why is this course important?

Figure III: Experimental Methods

A: RCTs

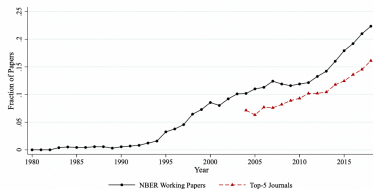


B: Lab Experiments

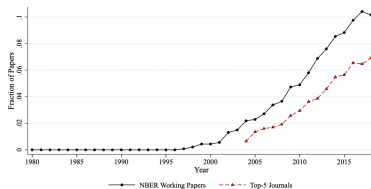
Why is this course important?

Figure IV: Quasi-Experimental Methods

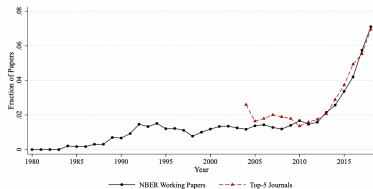
A: Difference-in-Differences



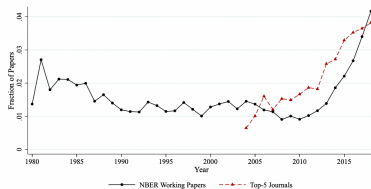
B: Regression Discontinuity



C: Event Study



D: Bunching



Notes: This figure shows the fraction of papers referring to each type of quasi-experimental approach. See Table A.1 for a list of terms. The series show 5-year moving averages.

More recent numbers

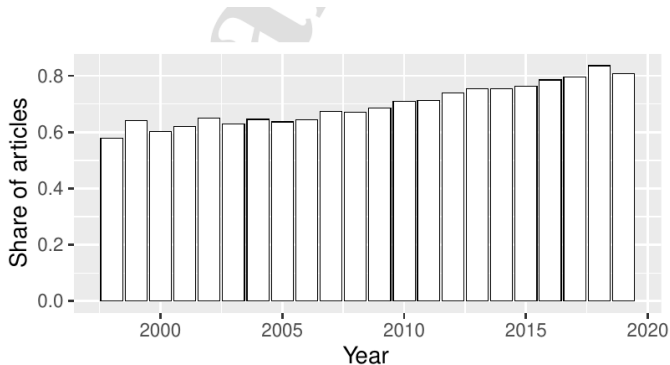
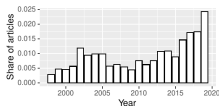


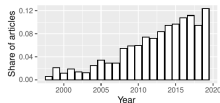
Figure 3: Share of empirical articles per year (all articles: 31k)

Brodeur et al. (2026), *European Economic Review*.

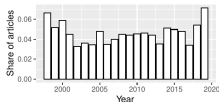
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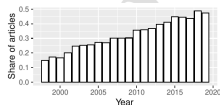
(a) Bunching



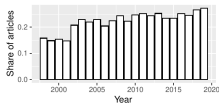
(b) Difference-in-differences



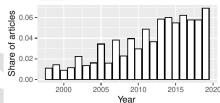
(c) Event study



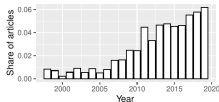
(d) Identification



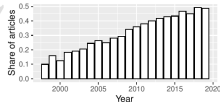
(e) Instrumental variables



(f) Lab experiment



(g) Field experiment



(h) Fixed effects

Economics is becoming more and more empirical, but...

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- But empirical work is hard
 - Fundamentally messy.
 - No clear start (statement of a theorem) and end (proof of the theorem).
 - Data is always messy $\rightarrow$ discretion will always come into play.
 - There is no Q.E.D at the end of empirical papers.

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This can be overwhelming for grad students entering their third year.

- They need to do research *independently*.
- Often this research is empirical.
- But most of your prior training was just about proving theorems...

Econ 628 is designed to smooth this transition

Pop-Quiz! From Dynarski et al. (2021, AER)

1740

THE AMERICAN ECONOMIC REVIEW

JUNE 2021

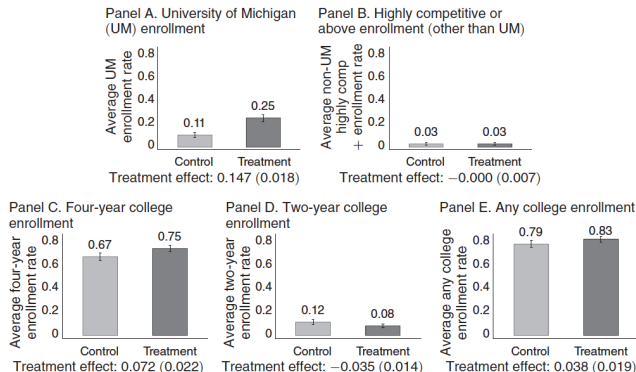


FIGURE 6. ESTIMATED EFFECT OF HAIL SCHOLARSHIP ON COLLEGE CHOICE, FIRST AND SECOND HAIL COHORTS

Notes: All analyses done at the school-year-level. Ninety-five percent confidence intervals shown based on standard errors clustered at the school level. Enrollment measured at the first institution attended in the fall following expected high school graduation. Enrollment unconditional on any college enrollment. Treatment effects estimated from a regression of the outcome on an indicator for treatment status and strata dummies. Robust standard errors clustered at the school level reported in parentheses.

Source: Michigan administrative data

One more... (this is from MHE)

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Table 3.2.1: Estimates of the returns to education for men in the NLSY

	(1)	(2)	(3)	(4)	(5)
Controls:	None	Age dummies	Col. (2) and additional controls*	Col. (3) and AFQT score	Col. (4), with occupation dummies
	0.132 (0.007)	0.131 (0.007)	0.114 (0.007)	0.087 (0.009)	0.066 (0.010)

Notes: Data are from the National Longitudinal Survey of Youth (1979 cohort, 2002 survey). The table reports the coefficient on years of schooling in a regression of log wages on years of schooling and the indicated controls. Standard errors are shown in parentheses. The sample is restricted to men and weighted by NLSY sampling weights. The sample size is 2434.

*Additional controls are mother's and father's years of schooling and dummy variables for race and Census region.

Last one... (this is from Borjas, 1980)

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TABLE 1
SELECTED WAGE COEFFICIENTS
DEPENDENT = $\ln(H_w)$

Measure of Wage (in logs)	Sample			
	Pooled	Hourly	Weekly	Annual
Usual wage rate	-.0383 (-2.37)	.0664 (1.51)	-.1040 (-3.81)	-.0256 (-1.11)
Predicted wage rate	.0269 (.59)	.1562 (1.47)	.0596 (.22)	.0863 (1.30)
Sample size	1561	278	449	834

Note: *t*-ratios are given in parentheses in all tables of this paper. Held constant in the regression are: other nonwage income in the family, expected number of years until retirement, years of labor force experience, number of children, whether job information refers to current or last (if not currently working) job, marital status, health status, and education.

Section 1

What is Econometrics?

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① Identification

② Estimation

③ Inference

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- ① Identification (What can be learned?)
 - ② Estimation (How do I learn it?)
 - ③ Inference (How certain should I be?)
- Econometricians: new results in these areas.
 - Applied Economists: formulate hypothesis, apply tools 1-3 created by econometricians to real data.

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- Good question. Some differences:
 - Applications to economic questions / data.
 - Emphasis on causal relationship using observational data.
 - Impose restrictions from economic theory.
 - Emphasis on welfare and counterfactual distributions.

Section 2

Econometric Models and Causality

Roadmap

- Today: Establish a link between basic concepts in econometric models and their relationship with modern notions of causality.
- For a long time in economics causality could be achieved/discussed only *after* having specified an econometric model.
- An econometric model must have two fundamental features:
 - It tells us how the observed data were generated.
 - It tells us how things would have been if we were to exogenously change one factor *ceteris paribus*
- Econometric models impose structure on the data generating process (DGP)
- Knowledge of the DGP yields a probability measure $\mathbf{P}(\cdot)$ over the observed data $\mathbf{Y}$.

- 1 Parametric models: models are indexed by a finite dimensional vector of parameters.
- 2 Non-parametric models: models are indexed by infinite dimensional parameters (e.g. an unknown function).
- 3 Semi-parametric models: models are indexed by a finite dimensional vector parameters and an infinite dimensional nuisance function.

A parametric model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad (1)$$

$$(X_i, \varepsilon_i) \sim_{\text{iid}} \mathcal{N} \left[\begin{pmatrix} \mu \\ 0 \end{pmatrix}, \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma^2 \end{pmatrix} \right] \quad (2)$$

$$\theta = (\beta_0, \beta_1, \mu, \log \sigma_x, \log \sigma) \in \mathcal{R}^5 \quad (3)$$

- Everything there is to know about the world in 5 parameters.

A semi-parametric model

$$Y_i = \beta X_i + \varepsilon_i \quad (4)$$

$$\mathbb{E}[\varepsilon_i | X_i] = 0 \quad (5)$$

$$\beta \in \mathcal{R} \quad (6)$$

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$$\beta \in \mathcal{R} \quad (6)$$

- Unrestricted distribution of $F_X(\cdot)$ of X_i .
- Mean restriction on conditional distribution of $F_{\varepsilon|X}$
- Both objects potentially infinite dimensional (but nuisance?)

A non-parametric model

$$Y_i = g(X_i, \varepsilon_i) \quad (7)$$

$$\varepsilon_i \perp X_i \quad (8)$$

$$g(., .) : \mathcal{R}^2 \rightarrow [0; 1] \text{ is monotone and increasing in both arguments} \quad (9)$$

- Unrestricted marginals of ε and X_i .
- Interested in function $g(., .)$ or features such as $E_\varepsilon[g(X_i, \varepsilon_i)]$.

Identification

- Identification is the study of which DGPs in the model space can be ruled out given the joint distribution of observables.
- This question is asked in world where DGPs can be estimated infinitely many times (that is never confuse identification with estimation).
- Logical first step for any empirical project.
- If nothing is identified, not worth going into step 2 and 3 (estimation, inference).

- Most models can be written as

$$\mathbf{Y}_i = \mathbf{g}(\mathbf{U}_i), \quad \mathbf{U}_i \sim_{iid} F_U(\mathbf{u}) \quad (10)$$

- We call $\theta \equiv (\mathbf{g}(\mathbf{U}_i), F_U)$ the structure of the model.
- $F_Y(\mathbf{y})$ the distribution function governing the observed variables $\rightarrow$ e.g. what is observed mean Y in my data.
- $F_\theta(\mathbf{y})$ the distribution function implied by θ under the model $\rightarrow$ e.g. what is the mean Y for a particular configuration of θ .

Setup

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- $F_\theta(\mathbf{y})$ the distribution function implied by θ under the model $\rightarrow$ e.g. what is the mean Y for a particular configuration of θ .
- Two structures θ' and θ'' are observationally equivalent if $F_{\theta'}(.) = F_{\theta''}(.)$

- The identified set is given by

$$\Omega(F_Y, \Theta) = \{\theta \in \Theta : F_\theta(.) = F_Y(.)\} \quad (11)$$

- Structure is point identified is $\Omega(F_Y, \Theta)$ is a singleton.

Example: A Mixture Model

Consider the model

$$Y_i = (1 + D_i) \varepsilon_i, \quad (D_i, \varepsilon_i) \stackrel{iid}{\sim} \text{Bernoulli}(p) \times N(0, 1), \quad p \in (0, 1)$$

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Proof.

By law of total probability:

$$f_Y(y) = \phi(y) (1 - p) + \frac{1}{2} \phi\left(\frac{y}{2}\right) p$$

Solving for p , we have $p = \frac{f_Y(y) - \phi(y)}{\frac{1}{2}\phi\left(\frac{y}{2}\right) - \phi(y)}$



- How many moments in the data can help you identify p ?

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- This question is the foundation of over-identification tests. We will go back to this concept many times!

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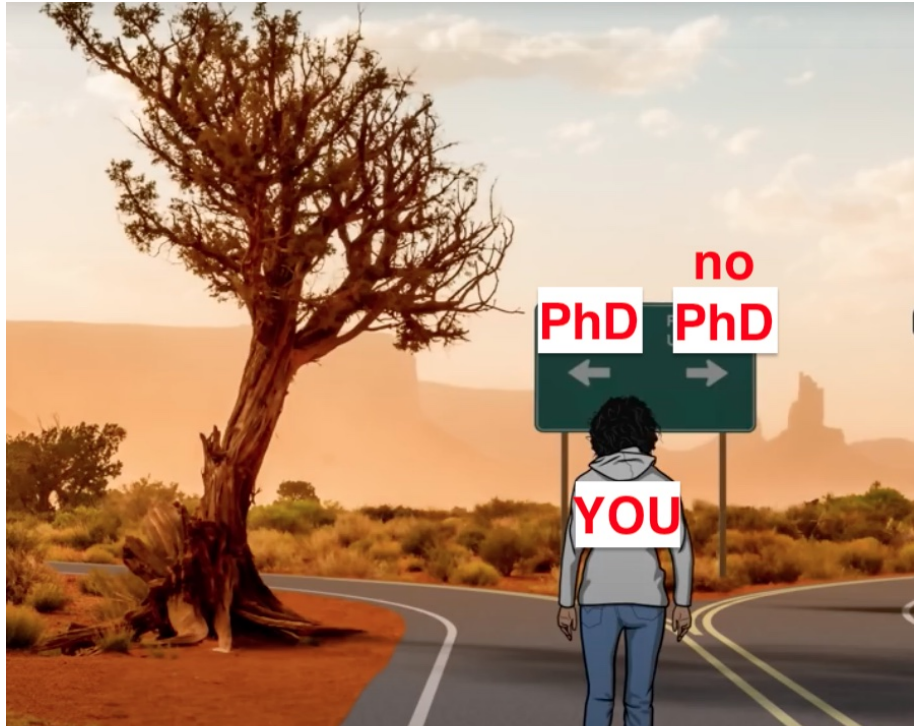
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- Causal effect of changing X_i from x' to x'' :

$$\Delta_i(x'', x') \equiv f(x'', U_i; \phi) - f(x', U_i; \phi)$$

- Need to be careful: can we change X_i holding $f(., .)$ and U_i constant?
- More generally, what if my model is wrong?



Potential Outcomes

- Statistics approach (Neyman-Rubin): Define counterfactuals “atheoretically” relative to what would emerge from an experiment.
- Potential outcome $Y_i^d \equiv Y_i(d)$ gives what results when unit i is experimentally exposed to treatment d

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- Example: Drug trial
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- “Fundamental problem of causal inference” (Holland, 1986)

$$Y_i = D_i Y_i^1 + (1 - D_i) Y_i^0$$

Unconfoundedness and ATE

i	Y_i^0	Y_i^1	D_i
1	32	20	1
2	200	4	0
3	0	0	0
4	80	80	1
5	53	88	0
6	0	90	1

- Causal inference requires restrictions on the “missing” potential outcomes
- Classic assumption: Unconfoundedness / Missing at Random

$$D_i \perp (Y_i^1, Y_i^0) \quad \forall i \quad (13)$$

- Under the above assumption we can identify

$$ATE = E[\Delta_i] = E[Y_i^1 - Y_i^0]$$

$$\begin{aligned} E[Y_i | D_i = 1] - E[Y_i | D_i = 0] &= E[Y_i^1 | D_i = 1] - E[Y_i^0 | D_i = 0] \\ &= E[Y_i^1] - E[Y_i^0] \\ &= E[Y_i^1 - Y_i^0] \\ &= ATE. \end{aligned} \quad (14)$$

What would OLS identify?

- We can express the potential outcome equation as follows

$$\begin{aligned} Y_i &= Y_i^0 + (Y_i^1 - Y_i^0)D_i \\ &= \mu + \beta D_i + \varepsilon_i \end{aligned} \tag{15}$$

where $\mu = E[Y_i^0]$; $\beta = E[Y_i^1 - Y_i^0]$ and
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- Does this look familiar?
- Just run OLS and you will get back ATE since randomization ensures that $E[D_i \varepsilon_i] = 0$
- Where is the randomness or uncertainty coming from here?

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- “Structural” econometrics literature relies on latent variable notation
 - combines restrictions on outcomes with restrictions on assignment
 - emphasizes “external validity” and welfare impacts
- Less assumptions better but can't get something for nothing
- Important to consider implicit assumptions present in both approaches..
- Successful JMPs combine both approaches (complements, not substitutes!).

SUTVA

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$$Y_i(d_1, d_2, \dots, d_i, \dots, d_N) = Y_i(d_i)$$

- Note that these assumptions are typically embedded in the $f(\cdot, \cdot)$ of a structural model:

$$Y_i = f(D_i, \varepsilon_i)$$

Which if's have causal answers?

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- Q1: What is the effect of being black (as opposed to white) on wages?

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- Large literature (e.g., Oaxaca, 1978) attempts to infer racial/gender discrimination from wage regressions
- Q1: What is the effect of being black (as opposed to white) on wages?
- Holland (1986): Q1 not well defined
 - Race is an “attribute” not a “treatment.” Can't be changed *ceteris paribus*.
 - “No Causation Without Manipulation!”

Which if's have causal answers?

- Large literature (e.g., Oaxaca, 1978) attempts to infer racial/gender discrimination from wage regressions
- Q1: What is the effect of being black (as opposed to white) on wages?
- Holland (1986): Q1 not well defined
 - Race is an “attribute” not a “treatment.” Can’t be changed *ceteris paribus*.
 - “No Causation Without Manipulation!”
- Q2: What is the effect of being perceived to be black (as opposed to white) on wages?
 - Perceptions can be manipulated (e.g. Bertrand and Mullianathan, 2004).
 - Explosion of “Audit Studies” → Another thing “invented” by Economists?

Audit Studies: Bertrand and Mullainathan (2004)

Are Emily and Greg More Employable than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination

Marianne Bertrand and Sendhil Mullainathan

NBER Working Paper No. 9873

July 2003

JEL No. J7, J71, J23, J24, J63, J82, C93

ABSTRACT

We perform a field experiment to measure racial discrimination in the labor market. We respond with fictitious resumes to help-wanted ads in Boston and Chicago newspapers. To manipulate perception of race, each resume is assigned either a very African American sounding name or a very White sounding name. The results show significant discrimination against African-American names: White names receive 50 percent more callbacks for interviews. We also find that race affects the benefits of a better resume. For White names, a higher quality resume elicits 30 percent more callbacks whereas for African Americans, it elicits a far smaller increase. Applicants living in better neighborhoods receive more callbacks but, interestingly, this effect does not differ by race. The amount of discrimination is uniform across occupations and industries. Federal contractors and employers who list “Equal Opportunity Employer” in their ad discriminate as much as other employers. We find little evidence that our results are driven by employers inferring something other than race, such as social class, from the names. These results suggest that racial discrimination is still a prominent feature of the labor market.

Blind Auditions: Goldin and Rouse (2000)

Orchestrating Impartiality: The Impact of
“Blind” Auditions on Female Musicians
Claudia Goldin and Cecilia Rouse
NBER Working Paper No. 5903
January 1997
JEL Nos. J7, J2, J4
Labor Studies

ABSTRACT

Discrimination against women has been alleged in hiring practices for many occupations, but it is extremely difficult to demonstrate sex-biased hiring. A change in the way symphony orchestras recruit musicians provides an unusual way to test for sex-biased hiring. To overcome possible biases in hiring, most orchestras revised their audition policies in the 1970s and 1980s. A major change involved the use of “blind” auditions with a “screen” to conceal the identity of the candidate from the jury. Female musicians in the top five symphony orchestras in the United States were less than 5% of all players in 1970 but are 25% today. We ask whether women were more likely to be advanced and/or hired with the use of “blind” auditions. Using data from actual auditions in an individual fixed-effects framework, we find that the screen increases — by 50% — the probability a woman will be advanced out of certain preliminary rounds. The screen also enhances, by severalfold, the likelihood a female contestant will be the winner in the final round. Using data on orchestra personnel, the switch to “blind” auditions can explain between 30% and 55% of the increase in the proportion female among new hires and between 25% and 46% of the increase in the percentage female in the orchestras since 1970.

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- Does this invalidate most of econometrics?
- tl;dr: No! We can impose structure to learn more (+ sometimes it is perfectly ok to only discuss effects of encouragement).
- More on this when we'll talk about IV.